

PAPER_457 — MUGE 38-System Extended Environment: F_torque + F_shock + F_cosmo Auto-Cascade

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 116 (v4.73) / Whitepapers created Session 121

Source: grok_share_e70525fa.txt (Doc 42 — MUGECompressed38System)

Classification: FIRST F_torque gravitational tidal torque term in UQFF; FIRST F_shock supernova/stellar-wind shock front term; FIRST auto-cascade from QG+DM+GW to F_cosmo

Author: Daniel T. Murphy

CP4 Class: MUGECompressed38SystemExtendedEnvCalculator (#95, PAPER_457)

Abstract

The 38-system MUGE extension expands the Session 116 29-system registry by adding 9 new object classes including the Lagoon Nebula, spiral supernovae host NGC-class systems, bipolar planetary nebula NGC 6302, and the full Orion region — while introducing two new fundamental environmental factor types: F_torque (gravitational tidal torque) and F_shock (supernova/wind shock). The auto-cascade mechanism automatically adds QG_term + DM_term + GW_term to F_cosmo for any system flagged with TYPE_UNIVERSE or TYPE_COSMOLOGICAL. Together, F_torque and F_shock bring the total F_env component count from 13 to 15, covering all identified astrophysical gravitational environments.

2. New Systems 30–38 — PAPER_457

2.1 Extended System Catalog (Systems 30–38)

#	System	Type	Key physics
30	LagoonNebula (M8)	HII region	O-star photo-ionisation, SFR
31	SpiralsSN	Spiral + SN	Shock injection into ISM
32	NGC6302	Bipolar PN	Tidal torque from binary WD
33	OrionNebula	HII	H-alpha resonance (see PAPER_447)
34	YoungStarsOutflow	YSO cluster	P_outflow (see PAPER_449)
35	EagleNebulaRepeat	HII	P_rad from NGC 6611
36	GravityBigBang	Cosmological	F_cosmo auto-cascade
37	WolfRayet151	Wolf-Rayet	Fast wind v_wind=2000 km/s
38	IC418 (Spirograph)	Compact PN	Dense ionised shell

2.2 F_torque — Gravitational Tidal Torque (FIRST in UQFF)

For binary systems and tidally interacting galaxies:

$$F_{\text{torque}} = \frac{GM_1M_2}{r_{12}^2} \cdot \frac{r}{r_{12}} \cdot \sin\left(\frac{\Omega_{\text{sync}} - \Omega_{\text{spin}}}{\Omega_{\text{sync}}}\right)$$

Simplified leading-order form used in the registry:

$$F_{\text{torque}} = 1 \times 10^{-11} \text{ m/s}^2 \quad [\text{canonical value for WD/PN systems}]$$

The torque term represents tidal dissipation rate — the transfer of orbital angular momentum of the binary into the envelope’s gravitational field. This is a **new gravitational coupling** that does not appear in any standard Newtonian or GR treatment.

Physical origin: In NGC 6302, the white dwarf binary separated at distance $d \sim \text{few AU}$ exerts differential tidal force on the bipolar lobes. The torque term captures the aspherical lobe-injection gravity.

2.3 F_shock — Supernova/Wind Shock Front (FIRST in UQFF)

For shock-driven bubble environments (SpiralsSN, WolfRayet151, IC443 SNR):

$$F_{\text{shock}} = \frac{\rho_{\text{post}} v_s^2}{r_{\text{shock}}} \cdot \delta(r - r_{\text{shock}})$$

Smoothed continuous form (avoiding delta function in numerical code):

$$F_{\text{shock}} = \frac{\rho_{\text{post}} v_s^2}{r} \cdot \exp\left(-\frac{(r - r_{\text{shock}})^2}{2\sigma_{\text{shock}}^2}\right)$$

Registry canonical value:

$$F_{\text{shock}} = 1 \times 10^{-11} \text{ m/s}^2 \quad [\text{at } r = r_{\text{shock}}]$$

This represents the gravitational acceleration produced by the **shell density enhancement** at the shock front — a term that couples the explosive dynamics of supernovae to the gravitational field.

2.4 Auto-Cascade: QG + DM + GW → F_cosmo

For systems flagged TYPE_UNIVERSE or TYPE_COSMOLOGICAL, the code auto-assembles F_cosmo:

```
if (system.type == UNIVERSE || system.type == COSMOLOGICAL):
    F_env += QG_term(t)      // h/l_p^2 * t/t_p * 1/(Mc^2)
    F_env += DM_term        // 0.268 * g_Newton
    F_env += GW_term(t)     // h_strain * c^2/_gw * sin(2 ct/_gw)
    // This cascade is F_cosmo
```

This is the **first automated cascade** in the UQFF codebase — eliminating manual F_cosmo assembly for cosmological systems.

3. Updated 15-Component F_{env}

Full updated F_{env} component table (15 terms):

#	Component	Formula	New in v38?
1–13	(as PAPER_456)	(as PAPER_456)	No
14	F_{torque}	$G M M_{\text{r}}/(r^3) \times \sin(\Delta /)$	NEW
15	F_{shock}	$_{\text{post}} v_{\text{s}}^2 \exp(-(r-r_{\text{sh}})^2/2^2)/r$	NEW

3.1 Wolf-Rayet Wind at $v_{\text{wind}} = 2000 \text{ km/s}$

WR 151 (WN4 star) has documented terminal wind velocity $v_{\text{wind}} = 2000 \text{ km/s} = 2 \times 10^6 \text{ m/s}$.
Wind term:

$$F_{\text{wind,WR}} = \rho_{\text{ISM}} v_{\text{wind}}^2 = 10^{-21} \times (2 \times 10^6)^2 = 4 \times 10^{-9} \text{ m/s}^2$$

This exceeds the Newtonian gravity of the WR star at OB-association separation ($\sim 1 \text{ pc} \sim 3 \times 10^4 \text{ m}$):

$$g_{\text{WR,Newton}} = \frac{GM_{\text{WR}}}{r^2} = \frac{6.674 \times 10^{-11} \times 2 \times 10^{31}}{(3 \times 10^4)^2} = 1.48 \times 10^{-12} \text{ m/s}^2$$

The WR wind term exceeds Newtonian gravity by **2700** $\times$ — confirming that stellar wind dynamics govern WR system gravitational environments.

4. Lagoon Nebula (M8) Parameters

Parameter	Value
M (Hourglass+HH36 region)	$\sim 5 \times 10^{33} \text{ kg}$ ($\sim 2500 M_{\odot}$ total)
r	$\sim 1 \times 10^4 \text{ m}$ ($\sim 10 \text{ ly}$ HII core)
z	0.0013 (1250 ly)
O-star (9 Sgr) luminosity	$\sim 2 \times 10^{32} \text{ W}$
SFR	$\sim 0.05 M_{\odot}/\text{yr}$

$$g_{\text{Lagoon,UQFF}} \approx g_{\text{Newton}} + P_{\text{rad,Lagoon}} = 3.33 \times 10^{-12} + \frac{2 \times 10^{32}}{4\pi(10^{17})^2 \times 3 \times 10^8} \approx 3.33 \times 10^{-12} + 1.77 \times 10^{-12} \approx 5.1 \times 10^{-12} \text{ m/s}^2$$

5. Standard Model Comparison

Feature	SM	UQFF PAPER_457
Tidal torque in gravity	External perturbation	F_torque as g modifier
Shock front coupling	Separate MHD	F_shock as g modifier
F_cosmo assembly	Manual per calculation	Auto-cascade for TYPE_UNIVERSE
38-system gravity	38 separate solvers	1 call with system registry

6. Testable Predictions

1. **NGC 6302 tidal torque:** $F_{\text{torque}} \sim 10^{11} \text{ m/s}^2$ — should produce a specific bipolar-lobe acceleration asymmetry detectable in proper-motion measurements of the nebular lobes.
2. **WR 151 wind dominance:** $F_{\text{wind}}/g_{\text{Newton}} \sim 2700\times$ — implies the WR bubble expands at $\sim v_{\text{wind}}$ rate, consistent with observed WR bubble diameters of $\sim 10 \text{ pc}$ over $\sim 0.1 \text{ Myr}$.
3. **Auto-cascade for cosmological systems:** Verified by running all 3 TYPE_UNIVERSE systems in the registry and confirming QG_term is always $< 10^{12} \times g_{\text{Newton}}$ — negligible but correctly non-zero.

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